

Cartographer 2D Mapping

Algorithm Installation

Taking ros2-humble as an example, enter the following command in the terminal to install,

```
# Core packages
sudo apt install ros-humble-cartographer ros-humble-cartographer-ros -y
# Supplementary packages
sudo apt install ros-humble-cartographer-rviz -y
sudo apt install ros-humble-nav2-map-server -y
```

Transplanting Source Code

Place the [cartographer Source Code] from the [Attachments] folder into the src directory of your workspace, then go back to the workspace directory and compile,

```
colcon build
```

Starting Mapping

Terminal 1: Enter the following command to start the lidar,

```
ros2 launch vanjee_lidar_sdk start.py
```

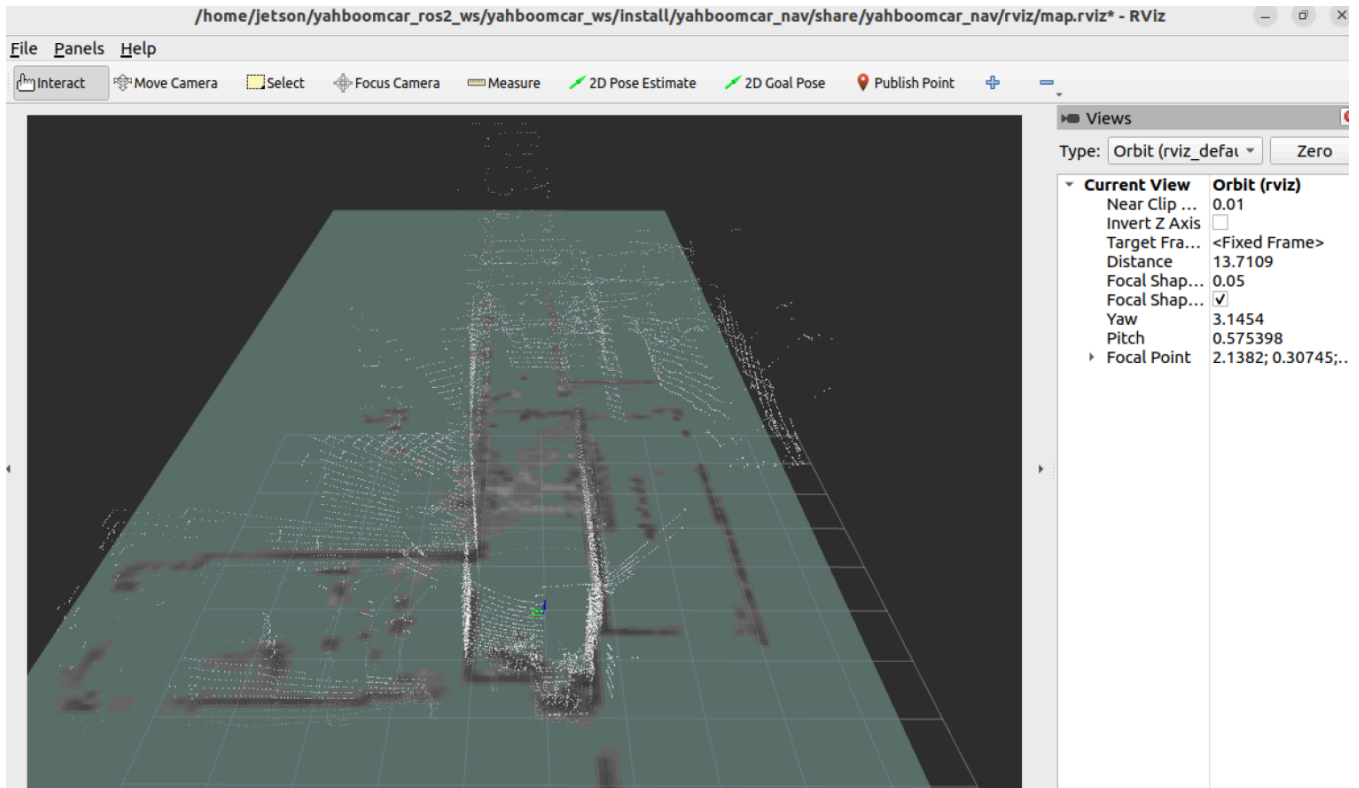
Terminal 2: Enter the following command to start rviz display,

```
ros2 launch yahboomcar_nav display_map_launch.py
```

Terminal 3: Enter the following command to start Cartographer mapping,

```
ros2 launch yahboomcar_nav cartographer_3d_launch.py
```

The rviz interface will display as follows,



Slowly move the lidar, the point cloud will move accordingly, and the grid map will update as well.

Saving the Map

Terminal 4: Enter the following command to save the map,

```
ros2 launch yahboomcar_nav save_map_launch.py
```

The saved map is a 2D grid map used for Nav2 navigation. The saved map is located in the directory,

```
yahboomcar_nav/maps/yahboomcar.pgm
```

Mapping Parameters

The cartographer mapping parameters are located in the yahboomcar_nav directory,

```
yahboomcar_nav/params/lds_3d.lua
```

For the parameter table as follows,

```
include "map_builder.lua"
include "trajectory_builder.lua"

options = {
  map_builder = MAP_BUILDER,
  trajectory_builder = TRAJECTORY_BUILDER,
  map_frame = "map",
  --Lidar coordinate system
  tracking_frame = "vanjee_lidar",
```

```

published_frame = "base_link",
odom_frame = "odom",
provide_odom_frame = true,
publish_frame_projected_to_2d = false,
use_odometry = false,
use_nav_sat = false,
use_landmarks = false,
num_laser_scans = 0,
num_multi_echo_laser_scans = 0,
num_subdivisions_per_laser_scan = 1,
num_point_clouds = 1,
lookup_transform_timeout_sec = 0.2,
submap_publish_period_sec = 0.3,
pose_publish_period_sec = 5e-3,
trajectory_publish_period_sec = 30e-3,
rangefinder_sampling_ratio = 1.,
odometry_sampling_ratio = 1.,
fixed_frame_pose_sampling_ratio = 1.,
imu_sampling_ratio = 1.,
landmarks_sampling_ratio = 1.,
}

```

```

MAP_BUILDER.use_trajectory_builder_2d = false
--Declare it is using 3D mode
MAP_BUILDER.use_trajectory_builder_3d = true
--Lidar detection distance
TRAJECTORY_BUILDER_3D.min_range = 0.3
TRAJECTORY_BUILDER_3D.max_range = 70.0
TRAJECTORY_BUILDER_3D.num_accumulated_range_data= 1
TRAJECTORY_BUILDER_3D.motion_filter.max_angle_radians = math.rad(1.0)
TRAJECTORY_BUILDER_3D.voxel_filter_size = 0.05
POSE_GRAPH.constraint_builder.min_score = 0.45
POSE_GRAPH.constraint_builder.global_localization_min_score = 0.45

--POSE_GRAPH.optimize_every_n_nodes = 30

return options

```